Arman Zareian

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Summary

Machine Learning Engineer, Data Scientist, and Full Stack Developer with 5+ years of experience building scalable ML and data systems across industry and academia. Deep expertise in privacy-preserving diffusion models, Large Language Models (LLMs), multi-agent architectures, GANs, and vision transformers. Graduate research focused on differential privacy for generative AI,multi-agent LLM inference, and privacy-utility optimization in Stable Diffusion variants. Proven track record in latency optimization, large-scale model deployment, identity obfuscation techniques, and end-to-end ML pipelines. Published in Artificial Intelligence Review (Q1, 60+ citations).

Technical Skills

- **Machine Learning:** PyTorch, Keras, scikit-learn, NumPy, Pandas; Vision Transformers; GANs; Diffusion Models
- **LLM / Agents:** LangGraph, LangChain, LlamaIndex; RAG; tool calling; MCP;
- **Visualization:** Matplotlib, Seaborn, Plotly
- **Backend / APIs:** Python, Django, FastAPI; Node.js, TypeScript; Express.js, Next.js; PHP (Laravel); REST APIs
- **Data / MLOps / Cloud:** Docker, Git, CI/CD, Airflow; Data Pipelines; Model Deployment; RabbitMQ; AWS, GCP, Azure
- **Databases:** PostgreSQL, MySQL, MongoDB
- **Automation / Scraping:** Selenium, Scrapy
- **Systems:** Linux, macOS, Windows
- **Hardware / IoT:** ESP32, UART, Device-to-Cloud Integration
- **Other:** MATLAB, R, IPFS
- **Leadership:** Communication, Team Leadership

Work Experience

Nexlife Inc.
Developer
5/2026-8/2026

AI/ML Software Intern
- Developing and testing AI/ML-powered software applications in a production environment;
- Working on ML pipeline development, LLM integration, and backend engineering;
- Supporting real-world software solutions through debugging, testing, and feature development;

Kansas State University
GRA/GTA
12/2024-present

Graduate Research Assistant
- Designed and implemented an IoT-to-web communication pipeline between ESP32-based devices and a Django backend;
- Developed UART-based data protocols and REST APIs for real-time logging, monitoring, and scalable log visualization;
- Built backend architecture ensuring reliability & data integrity across the embedded-to-web stack;
- Collaborated with hardware researchers on backend, networking, and system integration;

Graduate Teaching Assistant
- Led labs on pipelining, memory hierarchy, and MIPS programming; graded assignments;
- Mentored Agile student teams on workflows, code quality, and project development;

Pattern
Developer
1/2024-12/2024

Senior Back-End Developer
- Led development of FaceDancer (cross-platform identity sync), cutting identity management time by 40%;
- Integrated Ergo & Cardano smart contracts into Hummingbot, enabling DeFi automation;
- Deployed high-availability services with Docker/RabbitMQ/IPFS; achieved 99.9% uptime; cut request latency 30%;
- Built LLM-powered Google Sheets analytics add-on; reduced reporting effort 50%;

Westview Developer 1/2024-3/2024

Full-Stack Developer
- Built Django admin for inventory; cut stock discrepancies by 30%;
- Developed ML demand forecasting; reduced waste 20%;
- Automated order tracking; lowered errors 60%; enabled 5 new B2B clients;

Ergo Foundation Developer 11/2023-12/2024

Full-Stack Developer
- Tripled ErgoAuctionHouse throughput; improved UX;
- Built React front-ends & blockchain wallets; reduced checkout time 40%;
- Implemented IPFS thumbnail caching; cut load time 60%

Research Experience

Kansas State University 9/2025-present

Multi-Agent Reasoning and Stance Detection with Large Language Models
- Developing multi-agent LLM architectures for target-aware stance detection and contextual reasoning;
- Designing adaptive worker-allocation strategies to improve reasoning efficiency and classification performance;
- Developing modular prompt-optimization methods for robust stance detection across diverse targets;
- Conducting comparative evaluations of LLM reasoning, aggregation, and prompt-optimization strategies;

Privacy Auditing and Privacy-Preserving Generative Models
- Developing statistical methods for measuring identity leakage and privacy guarantees in generative image models;
- Designing and evaluating identity-level privacy audits for diffusion models and face-swapping anonymization systems;
- Investigating noise injection, latent-space perturbation, and privacy–utility trade-offs across image-generation settings;
- Extending privacy-preserving learning methods to scarce medical-video data;

Multimodal Facial Representation Learning
- Developing multi-representation facial feature-fusion methods for cross-domain affect and distress recognition;
- Evaluating representation robustness under domain shifts and limited-data settings;

Isfahan University of Technology 10/2020-1/2025

Non-Regret Learning for Multi-Generative GANs
- Applied game-theoretic learning to stabilize GAN training; improved classification robustness;
- Published: Games of GANs: Game-Theoretical Models for GANs, Artificial Intelligence Review (Q1, IF 15.5, 82 citations) [Springer] [Google Scholar];

Education

Kansas State University 12/2024-present

Ph.D. in Computer Science
GPA: 4/4

Kansas State University 12/2024-08/2026

M.S. in Computer Science
GPA: 4/4

Isfahan University of Technology 2017-2022

B.S. in Artificial Intelligence & Robotics
GPA: 3.98/4
Top 5 students

Isfahan University of Technology **B.S. in Applied Mathematics**
2019-2022 Second Major
GPA: 3.98/4
Top 5 students

Publications & Awards

Conference Papers
- V. Bondalakunta, **A. Zareian,** et al., "Toward Interpretable Privacy Guarantees in Face-Swapping Anonymization," arXiv:2608.25750.
  IEEE Symposium on Security and Privacy (S&P), 2027;
- M. A. B. Shah, **A. Zareian,** et al., "Multi-Representation Facial Feature Fusion for Cross-Domain Infant Distress Recognition." 38th IEEE International Conference on Tools with Artificial Intelligence (ICTAI), 2026*;*

Under Review
- V. Bondalakunta, **A. Zareian,** et al., "A Privacy-Preserving Framework for Sharing and Learning from Scarce Medical Video."
  Poster paper submitted to HealthSec'26, Cybersecurity in Healthcare Workshop;
- **A. Zareian,** et al., "Picture the Epsilon: Pursuing Identity-Level Privacy Guarantees for Images." arXiv:2608.17147.
  Manuscript submitted to PoPETs;
- N. Haque, V. Bondalakunta, G. Amariucai, MA. Bin Shah, **A. Zareian,** et al., "Swapped Face, Same Person: Measuring the Identity That Face Swaps Leave Behind."
  Manuscript submitted to ICLR 2027;

Preprints & Manuscripts
- M. Sabbaghan, **A. Zareian,** et al., "Multi-Agent Reasoning with Adaptive Worker Allocation for Stance Detection." arXiv:2606.11609.
  Manuscript ready for submission to NAACL;
- M. Sabbaghan, **A. Zareian,** et al., "MoPrO-SD: Modular Prompt Optimization for Stance Detection." arXiv:2610.13386.
  Manuscript ready for submission to NAACL;

Journals Articles
- M.Moghaddam, **A.Zareian**, et. al., "Games of GANs: Game-Theoretical Models for Generative Adversarial Networks", Artificial Intelligence Review ,Volume 56, pages 9771–9807, (2023) (Impact Factor: 13.9);

Award
- Competitive entrepreneurship award for MANGT 840: Advanced Entrepreneurship, Fall 2025;

Peer Review
- IEEE Signal Processing Letters, Apr 2025;
- IEEE Signal Processing Letters, Jan 2025;

Licenses & Certifications

Anthropic
- Introduction to Subagents, May 2026;
- Introduction to Agent Skills, May 2026;
- Model Context Protocol: Advanced Topics, May 2026;
- Introduction to Model Context Protocol, May 2026;
- Claude with the Anthropic API, May 2026;

Hugging Face
- LLM Agents, May 2025;